COLORADO
Chief Information Officer &
Executive Director
Governor's Office of Information Technology

TS-INF-009: Infrastructure as Code Standard

Document ID	TS-INF-009
Title	Infrastructure as Code Standard
Executive Sponsor	Infrastructure Operations/Cloud Ops
Issuing Authority	Office of Information Technology (OIT)
Applies to	OIT and OIT-supported agencies, employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders.
Tier	Standard
Version	3.0
Effective Date	08/02/2023
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1. Purpose

This standard enables Infrastructure Operations (IO) to consistently automate the deployment and maintenance of infrastructure in a secure, efficient, and repeatable manner, providing specific requirements for automating deployments of infrastructure on-premises and in the cloud through an Infrastructure-as-Code (IaC) approach.

The ultimate purpose of this standard is to enable OIT to efficiently define and deploy infrastructure — such as networks, virtual machines, load balancers, security appliances, and connection topologies — so an IaC model generates the same environment every time it deploys, provisioning the infrastructure service request in a secure, consistent, effective, and repeatable manner through automated scripts.

2. Scope

These standards apply to OIT and OIT-supported agencies. This standard shall be used by employees of OIT, agencies supported by OIT, and vendors contracted by OIT or its stakeholders, as a means to enforce technology standardization and assist in making the usage, support, and/or purchase of technologies more consistent and efficient.

Standards instituted for use by approval of the OIT Architecture Review Board (ARB) govern purchase and deployment decisions deemed "go-forward" or "net-new", and are not intended as a retroactively enforceable mandate for immediate conformance, especially where cost implications may be prohibitive.

3. Roles and Responsibilities

Enterprise Architecture - Chief Enterprise Architect

- Chairs and conducts business of the Architecture Review Board (ARB).
- Updates effective date and/or last review date.
- Publishes the ARB-approved Standard to the OIT public-facing website.

Architecture Review Board (ARB)

- Engages stakeholders and subject matter experts to determine the standard.
- Reviews and approves the Standard (this document).

Infrastructure Operations/Cloud Ops

- Owner of Standard; engages stakeholders and subject matter experts to define the standard.
- Updates document standards and version number; submits/presents to the ARB for approval.
- Provides resources necessary to implement and support IaC systems in the state cloud environment.
- Selects IaC languages, submits selections to the governing committee, and develops templates that meet OIT standards.

Infrastructure Operations/Network Services

- Provides resources necessary to implement and support network components of IaC.

Azure DevOps Administration Team

- Provisions repository and pipeline spaces for both operations teams and project teams using IaC.

All OIT staff are responsible for being aware of the existence of this (and all) standard(s) and its purposes of authority and governance in support of OIT's statutorily mandated mission in service of Colorado State Government affairs.

4. Standards

Infrastructure deployed after establishment of this standard is governed by conceptual IaC architecture diagrams and a basic IaC flow maintained by Infrastructure Operations. The following IaC requirements apply:

- Infrastructure deployed after the establishment of this standard shall be deployed using IaC whenever possible.
- IaC language selection shall be performed by OIT infrastructure teams and accepted by the committee governing the platform.
- Evaluate, select, standardize, and train on IaC tools such as Terraform.
- Application and infrastructure configurations shall be stored in appropriate file types in organizationally managed repositories.
- Project-specific IaC code should be stored in a project-specific repository and deployed via a deployment pipeline.
- IaC code should be scanned in a repeatable manner; the code scanning and code review processes should be documented in Standard Operating Procedures (SOPs).
- IaC templates should be developed by or approved by platform management teams prior to dissemination to project teams and vendors.

- Ensure IaC files are scanned for security, compliance, and quality, with any findings addressed prior to deployment.
- Enable IaC scanning tools such as Checkov or Veracode to produce safe infrastructure code.
- Restrict access to the IaC pipelines and repositories to project personnel and infrastructure teams.
- Infrastructure configured using IaC cannot be modified manually via the console or API, since those changes create drift that will be overwritten by the next IaC deployment.
- Application IaC should be configured to deliver as close to identical infrastructure between development, test, and production environments as possible.

5. Variance and Exception Process

Any request to procure or deploy technology that is a variance to the standards and provisions stated within this document must be requested of, and approved by, the "Owner of Standard" listed under Roles and Responsibilities.

Requests may be acquired by completing the "EXCEPTION REQUEST to Technical Standard" form and including the completed form (with all signatures) along with other documentation required as part of standard procedures when executing a technology procurement or service deployment.

6. References

- NIST Special Publication 800-204, Security Strategies for Microservices-based Application Systems
- Microsoft: What is Infrastructure as Code (IaC)?
- HashiCorp: Automate Infrastructure on Any Cloud
- AWS: Infrastructure as Code
- Snyk: IaC security for devs and DevOps

7. Definitions

Infrastructure as Code (IaC): Provisioning infrastructure through automated scripts to achieve consistent, predictable, and repeatable configuration deployments.

Terraform: An open-source IaC tool for provisioning and managing cloud infrastructure, codifying infrastructure in configuration files that describe the desired state for a topology.

Declarative Definition Files: Files describing the components and configuration an environment requires, but not necessarily how to achieve that configuration.

Infrastructure Operations (IOs): Includes server, network, NetSec, and cloud teams.

Exception request: A document requesting an exception to a recognized standard, requiring a business or technical justification, subject to review and approval by an authority in the area.